AMAZON KINDLE STORE REVIEW SETIMENT ANALYSIS

By Olasunkanmi David Adisa

100598874



AGENDA

- Problems
- Purpose
- The data
- Insights
- Model and Results
- Solution
- Future Improvement

PROBLEMS

Increase demand in e-commerce

Voice of customer concern

Importance of Customer experience

Fail to meet expectation = lose money

Easy to buy and read with Kindle

INTRODUCTION

E-commerce demand is growing by the day. Because customers buy things without seeing or feeling them, customer voice ideas such as reviews and customer experiences are becoming increasingly significant in online retailers. If the organisation fails to address this client demand, it loses money due to a lack of strategic judgments.

This documenter chooses the dataset "Amazon Kindle Shop Reviews" and want to do an emotional analysis of the Amazon Kindle store reviews using Python. I will detect Good and Bad ratings using four different models, including Logistic Regression, SVC, and Random Forest, as well as three distinct algorithms, including Accuracy, AUC score, and Precision with one diplomatic learning which is LSTM (Long short-term memory networks)



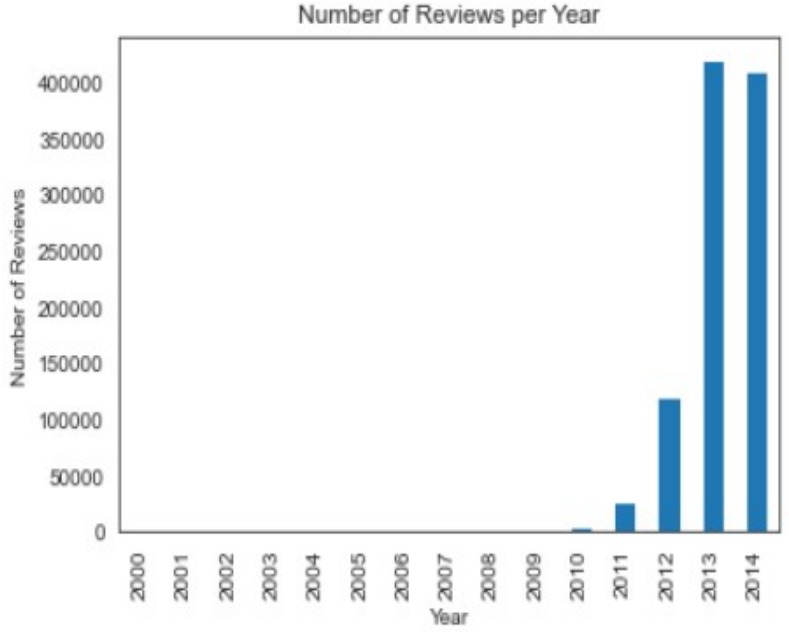
PURPOSE

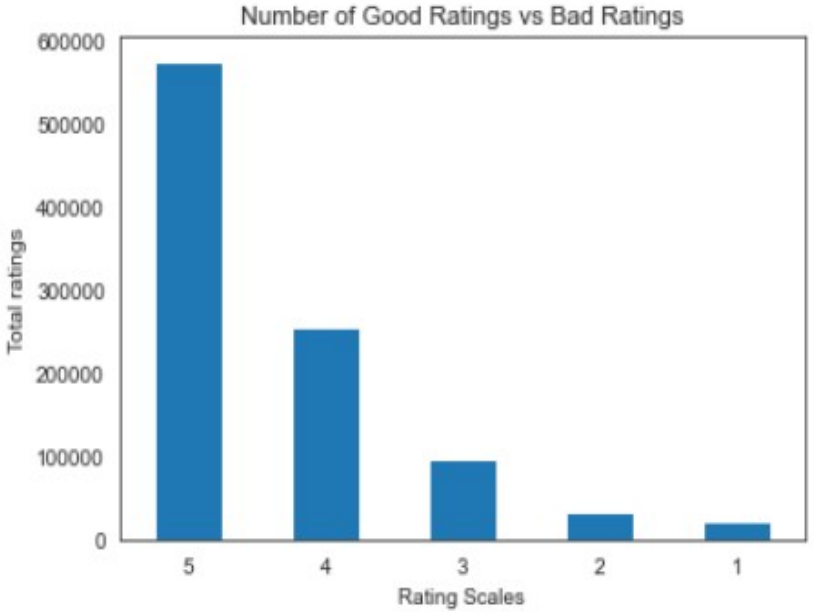
To readily identify evaluations as Good or bad rating in order to assist companies boost customer contentment and value their remarks in order to save the organisation money.

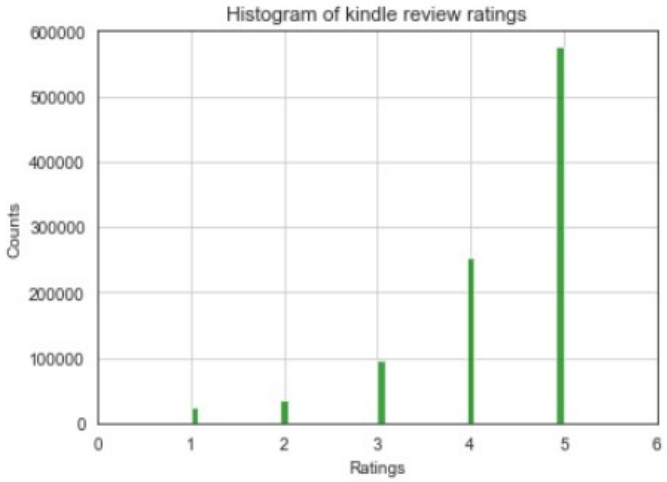
THE DATA

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 982619 entries, 0 to 982618
Data columns (total 10 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   Unnamed: 0      982619 non-null  int64
 1   asin            982619 non-null  object
 2   helpful         982619 non-null  object
 3   overall         982619 non-null  int64
 4   reviewText      982597 non-null  object
 5   reviewTime      982619 non-null  object
 6   reviewerID      982619 non-null  object
 7   reviewerName    978803 non-null  object
 8   summary         982618 non-null  object
 9   unixReviewTime  982619 non-null  int64
dtypes: int64(3), object(7)
memory usage: 75.0+ MB
```

EXPLORATORY DATA ANALYSIS








```
# checking missing values
df.isnull().sum()
```

```
Unnamed: 0          0
asin                0
helpful             0
overall             0
reviewText         22
reviewTime          0
reviewerID          0
reviewerName     3816
summary             1
unixReviewTime      0
dtype: int64
```

TEXT PRE-PROCESSING

- Removed punctuations
- Removed non alphabetical words
- Lowercased all words
- Removed stopwords
- Lemmatize to reduce words to their base form

```
rev.overall.value_counts()
```

```
1    925449
0     57148
Name: overall, dtype: int64
```

```
#testing the model with CountVectorizer
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = .3, random_state = 42, stratify = y)

naive = MultinomialNB()
naive.fit(X_train, y_train)
print('Training Accuracy w/ CountVectorizer: {:.2f}'.format(naive.score(X_train, y_train)))
print('Testing Accuracy w/ CountVectorizer: {:.2f}'.format(naive.score(X_test, y_test)))
```

```
Training Accuracy w/ CountVectorizer: 0.91
Testing Accuracy w/ CountVectorizer: 0.87
```

```
X.shape
```

```
(40000, 84510)
```

```
X_train.shape
```

```
(28000, 3690)
```

Machine learning

Naive Bayes

Accuracy: 87%
Precision: .87
Recall: .87
AUC Score: .94

Linear SVC

Accuracy: 87%
Precision: .87
Recall: .87
AUC Score: .87

Logistic Regression

Acuracy: 86%
Precision: .86
Recall: .86
AUC Score: .94

Random Forest

Accuracy: 85%
Precision: .85
Recall: .85
AUC Score: .92

Neural network

```
Model: "sequential"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
embedding (Embedding)        (None, 200, 32)           27066272

spatial_dropout1d (SpatialD  (None, 200, 32)           0
ropout1D)

lstm (LSTM)                  (None, 50)                16600

dropout (Dropout)            (None, 50)                0

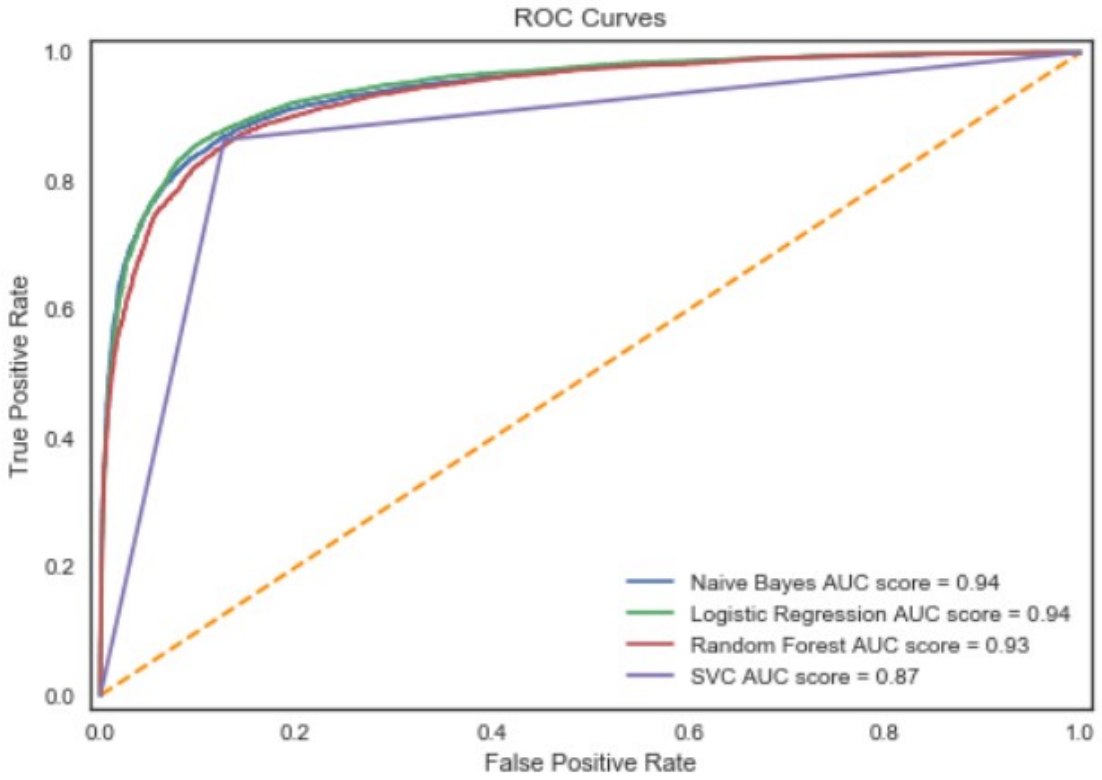
dense (Dense)                (None, 1)                 51

=================================================================
Total params: 27,082,923
Trainable params: 27,082,923
Non-trainable params: 0
_________________________________________________________________
None
```

```
history = model.fit(padded_sequence,sentiment_label[0],validation_split=0.25,epochs=3,batch_size=64)
```

```
Epoch 1/3
11515/11515 [==============================] - 7600s 658ms/step - loss: 9.8582e-04 - accuracy: 0.9999 - val_loss: 4.0198 - v
al_accuracy: 0.7674
Epoch 2/3
11515/11515 [==============================] - 17904s 2s/step - loss: 1.6266e-08 - accuracy: 1.0000 - val_loss: 5.2556 - val
_accuracy: 0.7674
Epoch 3/3
11515/11515 [==============================] - 87565s 8s/step - loss: 3.8107e-10 - accuracy: 1.0000 - val_loss: 5.6534 - val
_accuracy: 0.7674
```

ROC CURVES FOR MACHINE LEARNING



ADVANTAGES

1. EASY PRODUCT COMPARISON
2. DEFINING DISLIKES QUICKLY
3. SAVING TIME
4. MORE SALES
5. HAPPIER CUSTOMERS
6. LESS TIME ON SERVER = FEWER PROBLEMS

FUTURE IMPROVEMENT

- Expanded to do multi-class categorization of reviews

THANK YOU

